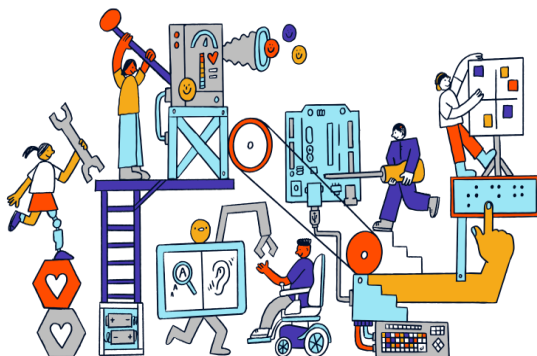


Product Manual: Adapted Projector Lamp Toy



Revision: 1.0

Date: 01-Dec-2025

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1. Introduction

1.1 About the Community Partner

Our journey began in 1970 with the will of Mrs Shakuntala Bhatia. Under the banner of the Asian Women's Welfare Association, Mrs Bhatia and her peers, like Mrs Tambyah and Mrs Kula, identified gaps in the community to help the underserved.

The Association incorporated AWWA Ltd on 7 January 2015, a Singapore company limited by guarantee. In April of the same year, AWWA Ltd took over the operations and activities previously managed by the Association. In 2022, the Association was dissolved, and AWWA continues to address social service gaps as they arise, by being guided by our mission and strategic vision.



1.2 Product Description

This table clarifies the intended purpose and scope of the product:

WHY?	Special needs students are often excluded from using toys and/or face lack of appropriate ways to express themselves.
WHO?	For special needs students who are learning to understand cause-and-effect and have fun.
WHAT?	A toy lamp with colourful light and projector star and moon onto the wall and ceiling. Adapted with an additional 3.5mm female mono audio cable to trigger power on/off by other switches.
HOW?	This product transforms an off-the-shelf projection lamp into a durable and responsive toy with a 3.5mm female audio jack so it can be triggered by another adapted switch or button. . It allows students to experience and understand the cause-and-effect.

1.3 Product Overview and Features



1. Adapted Projector Lamp Toy can be connected to an adapted recorder button and turn on/off with audio playback.
2. Adapted Projector Lamp Toy can be connected to an adapted limit switch and turn on/off.
3. Adapted Projector Lamp Toy can be connected to a squeeze or grip switch and turn on/off.

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1.4 Bill of Materials (BOM)

Item	Supplier/Product Link	Price	Quantity
Projection Lamp	Click Here -Link too long	6.62 SGD/pcs	1
3.5mm female mono audio jack cable	Click Here - Link too long	2.10 SGD/pcs	1
Cable Tie 3x100mm	Click Here - Link too long	1.62 SGD/100pcs	1

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2. User Guide: Instructions for Use



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2.1 Set Up Instructions

Use the following table for a step-by-step guide to setting up the product:

Setting up	
Open the battery cover and install 3 AA or 4 AAA batteries. Different models of the toy using different batteries.	
Connect it to an adapted button or switch to turn on/off it.	

Setting up

Remove the white cover to let it project to the wall and ceiling.



There are 3 modes on the projector lamp toy. **Current adaptation only support Mode B**
Mode A - Light up white LED and off
Mode B - Light up RED, BLUE, GREEN, RED BLUE, GREEN BLUE, RED GREEN, RED BLUE GREEN, pulse 3 LEDs and cycle through all 3 LEDs and off between it
Mode C - Rotation on and off



2.2 Maintenance

1. Remove AA batteries if the Adapted Projector Lamp Toy is not in use to prevent battery leak and corrosion.
2. If AA battery leaks, please change it
3. The device can be wiped clean with a damp cloth. Do not wash or submerge in water.

2.3 Troubleshooting

Problem	Possible Cause	What You Can Try
Projector Lamp toy not working	Battery drain	Change the battery to see if it works
	Battery spoilt because of leakage	Change the battery to see if it works
	Button is malfunctioning because battery leakage has caused the contact points to corrode	Check how bad the corrosion is? 1. If is very bad that it leak to the PCB , please replace the button, 2. If not try to clean up and replace the battery
	Projector Lamp toy faulty because of the solder broken	Open with the assembly instruction and try to check on it or contact EG for support.

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3. Maker Guide: Assembly Instructions



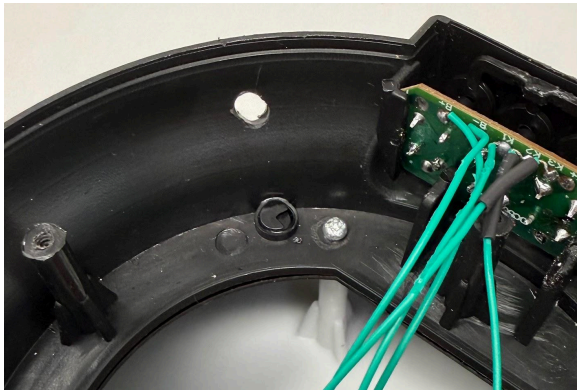
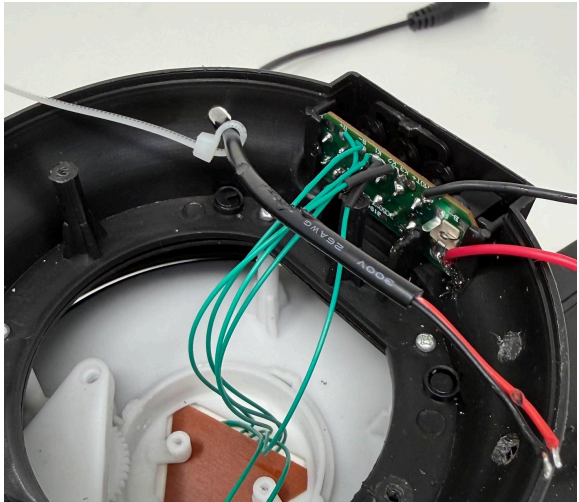
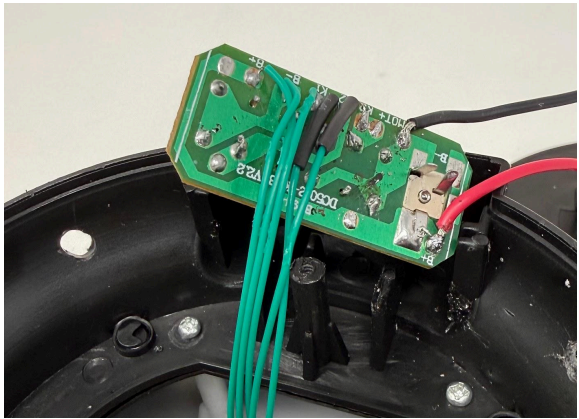
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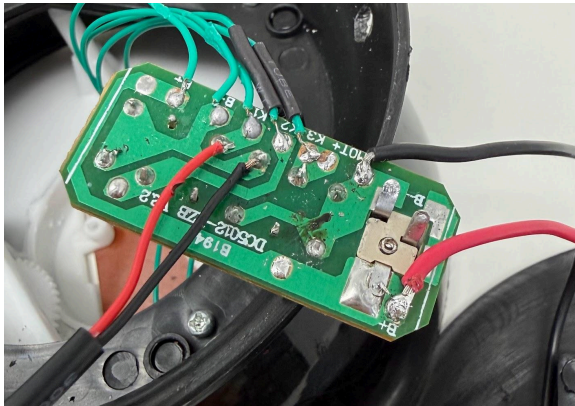
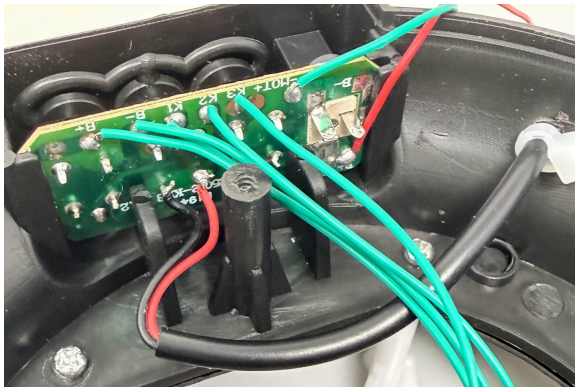
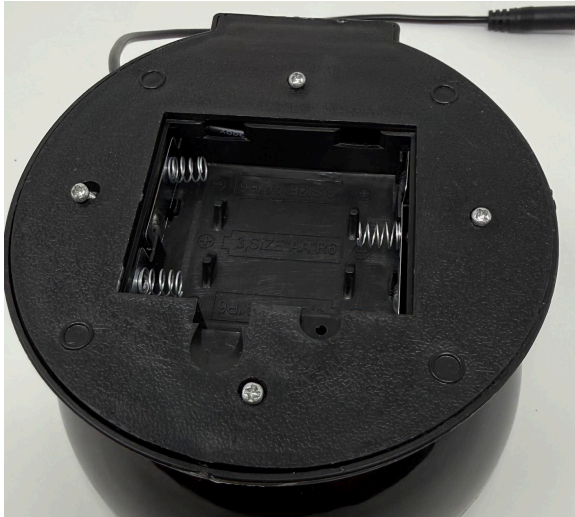
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3.1 Assembly Guide

Steps	Images
Remove the top white and black cover.	
Remove the battery cover.	

Steps	Images
Remove the battery if there is and unscrew the 4 screws.	
Open up the bottom cover.	

Steps	Images
Drill a hole of 4-4.5mm depending on the size of your 3.5mm audio jack cable.	
Loop through the 3.5mm female audio jack cable and put a cable tie to it.	
Remove the PCB from the slot.	

Steps	Images
Solder the 2 wires from the 3.5mm female audio jack to the pcb middle button. You can solder all the 3 buttons too.	
Place back the PCB to the slot properly and don't break the solder.	
Screw back the 4 screws	

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Steps	Images
Install 3 AA batteries and test it with a switch or button.	

4. Credit and Open Hardware Licence

This section provides the required legal notices and attribution for the open hardware design used to create this product.

4.1. Open Hardware Attribution

The core design for this Adapted Protector Lamp Toy is based on the Switch Adapted Moon and Star Projector, an open-source assistive technology project.

- Original Designer: Makers Making Change (a program of Neil Squire).
- Original Copyright: Copyright (c) Neil Squire / Makers Making Change.
- Original Source Location: The original project source files are available online at:

<https://www.makersmakingchange.com/product/switch-adapted-moon-and-star-projector/01tJR000000694HYAQ>



The screenshot shows the Makers Making Change website. At the top is a navigation bar with the logo, 'Assistive Devices', 'Make', 'Engage', 'Learn', 'Clubs That Care', 'More', and a search bar. Below the navigation bar is a large section titled 'About Makers Making Change' with the subtitle 'Empowering people with disabilities since 1984'. Under this is a section titled 'Our Mission' which contains text about the organization's goals and a video player. The video player shows three people working on a project and has a 'Watch on YouTube' button.

Makers Making Change
Assistive Devices Make ▾ Engage ▾ Learn ▾ Clubs That Care ▾ More ▾ Search... 🔍

About Makers Making Change

Empowering people with disabilities since 1984

Our Mission

According to Statistics Canada, 80% of people living with a disability use an assistive device to increase their independence, and 27% need at least one more. Cost is often cited as the main barrier.

Volunteer-made assistive technology can help bridge the gap.

Neil Squire's **Makers Making Change** program connects people with disabilities to volunteers who can build affordable assistive technology. Our mission is to achieve economic and social inclusiveness for all people with disabilities.

Watch on  YouTube

<https://www.engineeringgood.org/bespoke-projects/>

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4.2. License Statement and Source Availability

The hardware design used in this product is licensed under the CERN Open Hardware Licence Version 2 – Weakly Reciprocal (CERN-OHL-W 2.0) or later. By distributing this product, we are obligated to make the complete design source available to you.

- Complete Source Availability: The complete design files (Source), including schematics, assembly instructions, and any modifications made by Engineering Good, are available free of charge at a permanent online location:
<https://github.com/Engineering-Good/T4G-Adapted-Projector-Lamp-Toy>
- Modification Notice: This version of the Adapted Projector Lamp Toy design was modified by Engineering Good to include the 3.5mm mono audio jack functionality for connecting to other switches. The full, modified source is available at the URL listed above.
- Full License Text: The complete legal text of the CERN Open Hardware Licence Version 2 – Weakly Reciprocal** follows this section.

4.3 CERN Open Hardware Licence

The full text of the license is available here: [CERN-OHL-W 2.0 Full Text](#)